

Roberta Beatrix Southwood

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Washington, DC

US Citizen

Portfolio: <https://roberta-southwood.github.io/>

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TECHNICAL SKILLS

Programming Languages:

Python (NumPy, pandas, SciPy, Matplotlib, Seaborn) • R (tidyverse, ggplot2, Shiny) • SQL (MySQL, PostgreSQL) • MATLAB

Tools & Technologies:

Git, GitHub • LaTeX • Jupyter Notebook (Anaconda) • APIs • Tableau • ArcGIS • Microsoft Office • Google Workspace

EDUCATION

American University, Washington, DC

December 2026

Masters of Science in Data Science, GPA: 4.0/4.0

Michigan State University, East Lansing, MI

May 2025

Bachelors of Science in Mathematics, Minor in Computational Mathematics, Science, and Engineering (CMSE), GPA: 3.5/4.0

DATA SCIENCE EXPERIENCE

Data Science Intern

National Aeronautics and Space Administration (NASA) Headquarters, *June - August 2026*

- Analyzed 10 years of demographic data on NASA Announcement of Opportunity(AO) research proposal teams, identifying trends and insights aligned with Science Mission Directorate (SMD) leadership priorities.
- Developed accessible visualizations and statistical analyses for internal and external research publications, including comprehensive alt text.
- Collaborated with the SMD Data Science team and presented findings to NASA Headquarters SMD and leadership.
- Built reproducible data workflows and automation to streamline analysis replication and future updates.

Graduate Researcher (Data Science)

American University, *August 2025 - May 2026*

- Designed a data-driven research project applying sheaf modeling and topological methods to ecosystem analysis.
- Built a Python predictive model of wolf-moose predator-prey dynamics using six decades of real-world data.
- Evaluated model performance and accuracy using statistical methods.
- Presented research at StatConnect 2026 and the American University Mathias Student Research Conference.

Undergraduate Researcher (Computational Data Modeling)

Virginia Institute of Marine Science, *June -August 2024*

- Modeled mercury transport in Galveston Bay during Hurricane Harvey using the Regional Ocean Modeling System.
 - Applied computational methods to run large-scale numerical simulations on HPC systems.
 - Processed and analyzed large-scale model output using Python, MATLAB, and NetCDF.
 - Presented findings at the American Geophysical Union conference and contributed to a published manuscript.
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ADDITIONAL WORK EXPERIENCE

Teaching Assistant (Statistical Programming in R, Data Science, BioStatistics)

American University, *January 2026 - Present*

- Assist students in analyzing data using R reinforcing concepts in data cleaning, debugging scripts, exploratory data analysis, and reproducible analysis.
- Grade assignments and provide detailed feedback on statistical reasoning, code quality, and interpretation of results.
- Hold office hours to support students, breaking down complex statistical concepts into clear, actionable steps.

Undergraduate Learning Assistant (Calculus One)

Michigan State University, *August 2023 - May 2025*

- Improved student outcomes via data informed review sessions and targeted problem solving support.
- Consistently received top evaluation scores for communication and engagement.

Undergraduate Researcher

MSU Topology Summer Program, *May - July 2023*

- Translated mathematical theory into an analytical report and delivered technical presentations to diverse audiences.
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PUBLICATION

Du, Z., Harris, CK., Southwood, R., Yin, D., Bao, D., Ye, F., Dellapenna, TM., Xu, K., Xue ZG. (2026) Redistribution of Sediment and Particulate Mercury Within Upstream Reaches of Galveston Bay in Response to Hurricane Harvey. *JGR Oceans*, 131(8), <https://doi.org/10.1029/2025JC023581>